zadanie 1.

$$a\bar{b}c + (\bar{a}+b+d)(a+b\bar{d}+\bar{b})$$

$$(a+\bar{b}+c)\cdot(\bar{a}bd)+(a(b+\bar{a})\cdot\bar{b}) =$$

$$= a\bar{a}bd + \bar{a}b\bar{b}d + \bar{a}bcd + ab\bar{b} + a\bar{b}\bar{d} = \bar{a}bcd + a\bar{b}\bar{d}$$

funkcja dualna

$$a\bar{b}c + (\bar{a}+b+d)(a+b\bar{d}+\bar{b}) =$$

$$= a\bar{b}c + \bar{a}a + \bar{a}b\bar{d} + \bar{a}\bar{b} + ab + bb\bar{d} + b\bar{b} + ad + bd\bar{d} + \bar{b}d =$$

$$= a\bar{b}c + \bar{a}b\bar{d} + \bar{a}\bar{b} + ab + b\bar{d} + ad + \bar{b}d =$$

$$= a\bar{b}c + b\bar{d} + \bar{a}\bar{b} + ab + ad + \bar{b}d = ab + \bar{a}\bar{d} + \bar{b}d + c\bar{d} =$$

~~[illegible]~~

$$= [\text{illegible}] = abcd + abc\bar{d} + ab\bar{c}d + ab\bar{c}\bar{d} + \bar{a}bc\bar{d} + \bar{a}\bar{b}c\bar{d} +$$

$$\bar{a}\bar{b}c\bar{d} + \cancel{ab\bar{c}\bar{d}} + a\bar{b}c\bar{d} + \bar{a}\bar{b}\bar{c}\bar{d} + \bar{a}\bar{b}cd + a\bar{b}\bar{c}\bar{d} +$$

$$+ abc\bar{d} + \bar{a}\bar{b}c\bar{d} + \bar{a}bc\bar{d} + a\bar{b}\bar{c}\bar{d} \neq$$

$$= \cancel{abcd + abc\bar{d}}$$

Funkcja nie jest autodualna, ponieważ postacie dysjunkcyjne są inne.

zadanie 2.

$$\overline{a\bar{b}c + (\bar{a}+b+d)(a+b\bar{d}+\bar{b})} =$$

$$= \overline{a\bar{b}c} \cdot \overline{(\bar{a}+b+d)} + \overline{(a+b\bar{d}+\bar{b})} =$$

$$= (\bar{a}+\bar{b}+\bar{c}) \cdot (a\bar{b}\bar{d}) + (\bar{a}\cdot(\bar{b}+d)\cdot b) =$$

$$= (\bar{a}+b+\bar{c}) \cdot (a\bar{b}\bar{d}) + (\bar{a}b\bar{b} + \bar{a}bd) =$$

$$= (\bar{a}+b+\bar{c}) \cdot a\bar{b}\bar{d} + \bar{a}bd = a\bar{b}\bar{d} + \bar{a}bd$$

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zadanie 3

a) stała 0 i $f_1(a,b) = \bar{a}b + a\bar{b}$

1. stała jest
2. negacja: $f(a,a) = \bar{a}a + a\bar{a} = 0$
   $f(b,b) = \bar{b}b + b\bar{b} = 0$
   $f(a,0) = \bar{a}\cdot 0 + a\cdot 1 = a$
   $f(0,a) = 1\cdot a + 0\cdot\bar{a} = a$

nie można utworzyć negacji, więc zbiór funkcji nie jest zbiorem funkcjonalnie pełnym.

b) stała 1, $f_2(a,b) = \bar{a}b + a\bar{b}$

1. stała jest
2. negacja: $f(a,a) = \bar{a}a + a\bar{a} = 0$
   $f(b,b) = \bar{b}b + b\bar{b} = 0$
   $f(a,1) = \bar{a}\cdot 1 + a\cdot 0 = \bar{a}$ ✓

3. suma i iloczyn

$$f(f(a,1),b) = ab + \bar{a}\bar{b}$$
$$f(f(a,1),1) = a\cdot 1 + \bar{a}\cdot 0 = a$$

nie można utworzyć sumy lub iloczynu, więc zbiór funkcji nie jest zbiorem funkcjonalnie pełnym

zadanie 4

$$\bar{b}(\bar{a}+c)+b\bar{c} = \bar{a}\bar{b}+\bar{b}c+b\bar{c} =$$

$$= \bar{a}\bar{b}(c+\bar{c})+\bar{b}c(a+\bar{a})+b\bar{c}(a+\bar{a}) =$$

$$= \bar{a}\bar{b}c+\bar{a}\bar{b}\bar{c}+a\bar{b}c+\bar{a}\bar{b}c+ab\bar{c}+\bar{a}b\bar{c} =$$

$$= \underset{001}{\bar{a}\bar{b}c}+\underset{000}{\bar{a}\bar{b}\bar{c}}+\underset{101}{a\bar{b}c}+\underset{110}{ab\bar{c}}+\underset{010}{\bar{a}b\bar{c}}$$

$$f(a,b,c) = \Sigma m(0,1,2,5,6)$$

zadanie 5.

m	a b c d	X	Y
0	0 0 0 0	0	0
1	0 0 0 1	0	0
2	0 0 1 0	0	1
3	0 0 1 1	1	1
4	0 1 0 0	0	0
5	0 1 0 1	0	1
6	0 1 1 0	1	0
7	0 1 1 1	0	1
8	1 0 0 0	0	0
9	1 0 0 1	1	0
10	1 0 1 0	0	0
11	1 0 1 1	0	1
12	1 1 0 0	1	0
13	1 1 0 1	0	1
14	1 1 1 0	0	0
15	1 1 1 1	1	0

$$X - \Sigma m(3,6,9,12,15)$$

$$Y - \Pi m(0,1,4,6,8,9,10,12,14,15)$$

$$x = \bar{a}\bar{b}cd + \bar{a}bc\bar{d} + a\bar{b}\bar{c}d + ab\bar{c}\bar{d} + abcd$$

$$y = abcd + abc\bar{d} + a\bar{b}cd + a\bar{b}\bar{c}\bar{d} + \bar{a}bcd + \bar{a}bc\bar{d} + \bar{a}b\bar{c}\bar{d} + \bar{a}\bar{b}cd + \bar{a}\bar{b}c\bar{d} + \bar{a}\bar{b}\bar{c}d$$